

Assignment # 2 : Blockchain Technologies (SEN-604)

Subject: Blockchain Technologies (SEN-604)

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Total Marks: 10

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Mapped CLO: CLO 2 (Analyze blockchain platforms and evaluate their suitability).

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1. Introduction

Block chain is a digital ledger that stores records securely and makes them very hard to change.

There are three main types:

- **Public**
- **Private**
- **Consortium**

Each works differently and suits different needs. This assignment compares them and picks the best one for a voting system in Pakistan.

2. Comparison of Blockchain models

2.1 Public Blockchain

Open to everyone. Anyone can read, write, and validate (e.g., Bitcoin, Ethereum, Cardano, Polkadot, Solana.).

Transparency: High, all transactions visible.

Security: Strong due to decentralization, but 51% attacks possible.

Speed: Slow, around 7 transactions per second.

Scalability: Poor.

2.2 Private Blockchain

Controlled by one organization. Only authorized people can join (e.g., Hyperledger Fabric, Quorum, R3 Corda).

Transparency: Low, restricted access.

Security: High access control but single point of failure.

Speed: Fast, thousands per second.

Scalability: Excellent.

2.3 Consortium Blockchain

Shared by multiple organizations. Semi-decentralized (e.g., R3 Corda, Energy Web Chain, Quorum).

Transparency: Moderate, visible to members only.

Security: High, trust spread among known parties.

Speed: Fast, consensus among few trusted nodes.

Scalability: Very good.

2.4 Summary Table

Feature	Public	Private	Consortium
Access	Open to all	One organization	Multiple Organizations
Transparency	High	Low	Moderate
Security	High but risky	High But Centralized	High with Shared trust
Speed	Low	High	High
Scalability	Poor	Excellent	Very Good



3. Best Model for Voting in Pakistan

3.1 Why Consortium Blockchain is Suitable

A voting system needs security, privacy, transparency, speed, and trust. A consortium blockchain delivers all of these:

Shared control: ECP, political parties, and authorized observers can participate in validating and auditing the network, reducing dependence on a single authority.

Privacy with accountability: Votes stay private, but members can audit the ledger.

Fast and scalable: Handles millions of votes quickly.

Secure: Distributed validation stops tampering and fraud.

Auditable: Immutable records help settle disputes.

3.2 Why Not Public or Private

Public: Too slow, no voter privacy, open to attacks.

Private: Run by one authority, so other parties will not trust it.

4. Conclusion

The consortium blockchain is the best fit for Pakistan's voting system. It balances transparency, privacy, security, and speed better than public or private models.

5. Complete Diagram

